

Propagation of detonation waves

Y23 (2023) — English translation

Combustion is an exothermic chemical reaction (fuel + oxidizer), accompanied by positive specific heat release. Under certain conditions, it can be self-sustaining, in contrast to endothermic (accompanied by the absorption of heat) reactions. For example, when a source of ignition occurs in a pre-mixed gas mixture of fuel and oxidizer, a combustion wave can form. Detonation is a combustion mode in which a shock wave propagates through a substance, initiating chemical combustion reactions that support the movement of the shock wave.

A flame front is a narrow reaction zone of a propagating flame in which combustion occurs. This task asks you to study the propagation of detonation shock waves in a gas.

Let the gas move through a thermally insulated straight pipe of constant cross-section. The gas can be divided into two regions. In one region, the gas is motionless, its pressure is p_0 , density is ρ_0 . In another region, all elements of the gas move at the same speed, gas pressure p_1 , density ρ_1 . The speed of movement of the boundary between these regions c will be called the shock wave speed. The acceleration of free fall, as well as the thermal conductivity of the gas and its friction against the walls of the vessel can be neglected. Note: solving the problem will be significantly simplified if we reduce the gas movement to stationary. During stationary motion, the gas parameters at each point in space remain constant in time.

Part A. Shock wave in gas (3.0 points)

A wave in which sharp jumps in pressure and density occur near the wave front is called a shock wave. Let us consider a one-dimensional flow of an ideal gas with the adiabatic exponent γ . In this part of the problem, you are asked to analyze the relationships between the parameters p_0 , ρ_0 , p_1 and ρ_1 in the shock wave.

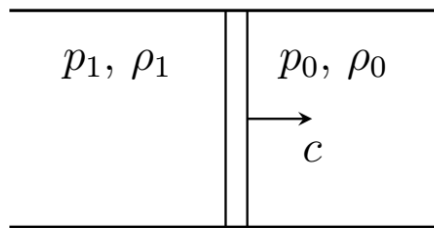


Figure 1: Figure 1.

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| A1 | Write down a system of laws of conservation of mass and momentum for the volume of gas passing through the wave front of the shock wave. Determine from it the speed of movement of the wave front c . | 1.00 |
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Since there are no dissipations in the system, the energy in it is also conserved.

A2 Write down the law of conservation of energy for an element of the volume of gas passing through the wave front. Obtain from it an additional equation relating the parameters p_0, ρ_0, p_1, ρ_1 , the speed of movement of the wave front c and the adiabatic exponent γ . **1.00**

A3 Combining the results obtained in steps A1 and A2, express the pressure ratio p_1/p_0 in terms of the adiabatic exponent γ and the compaction coefficient $k = \rho_1/\rho_0$. This relationship is called the adiabatic equation for a shock wave. **0.50**

A4 Determine the maximum possible value of the compaction coefficient k_{max} . Express your answer in terms of γ . Calculate the answer for an ideal diatomic gas. **0.50**

Part B. Propagation of detonation combustion wave (4.0 points)

When heated, some gases begin a chemical reaction and ignite. A shock wave can then propagate through the gas, and combustion occurs in a very thin layer near the wave front. In this part of the problem we will consider a detonation wave. The main difference between this part of the problem and part A is that when the mass Δm_0 enters the motion of the system (when this mass passes through the wave front), a chemical reaction occurs and heat $\Delta Q = q\Delta m_0$ is released, where q is a known quantity called the specific thermal effect of the chemical reaction.

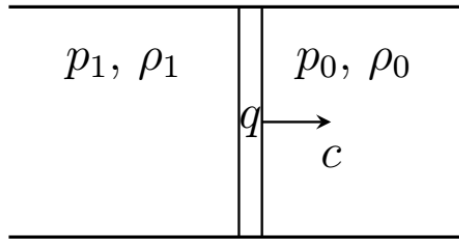


Figure 2: Figure 2.

Since during a chemical reaction the composition of the substance changes, the adiabatic index γ also changes. Let us denote by γ_0 and γ_1 the adiabatic exponents of the stationary gas and the gas after the passage of the wave front, respectively.

B1 Write down the law of conservation of energy for a detonation wave. Obtain from it an equation relating the parameters p_0, ρ_0, p_1, ρ_1 , the speed of movement of the wave front c , the specific thermal effect of the reaction q , as well as the adiabatic exponents γ_0 and γ_1 . **0.60**

Let us introduce the value of the specific volume $v = 1/\rho$ for each state: $v_0 = 1/\rho_0, v_1 = 1/\rho_1$. Denote the dependence $p_1(\rho_1)$ obtained at a fixed value of the shock wave speed c from the combination of the law of conservation of mass and the law of conservation of momentum as $p_1(\rho_1)_{imp}$. The dependence $p_1(\rho_1)$ obtained from the combination of the laws of conservation of mass, momentum, and energy will be denoted as $p_1(\rho_1)_{det}$. Next, work in the approximation of a strong detonation wave, i.e., a wave satisfying the conditions:

$$p_0 \ll p_1, \quad \frac{p_0}{\rho_0} \ll q.$$

B2 The explicit dependence $p_1(v_1)_{\text{det}}$ for a strong detonation wave can be represented as: $p_1 = \frac{q}{v_0} \cdot A$. Determine A . Express the answer in terms of γ_1 and the ratio v_1/v_0 . This relationship is called the detonation adiabatic equation. **0.60**

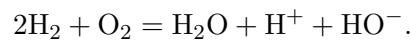
B3 On one graph, qualitatively depict the dependence $p_1(v_1)_{\text{det}}$, as well as the dependence $p_1(v_1)_{\text{imp}}$ for different values of the speed of movement of the wave front c . **0.60**

When studying detonation processes, Chapman and Jouguet hypothesized that the detonation combustion wave propagates at the minimum possible speed at which the equation $p_1(v_1)_{\text{imp}} = p_1(v_1)_{\text{det}}$ has a unique solution, called the Chapman-Jouguet point. This statement was later proven by Zeldovich and Doering.

B4 Determine the parameters p_1 and v_1 of the Chapman-Jouguet point. Express your answer in terms of v_0 , γ_1 , q . **1.40**

B5 Determine the speed of movement of the wave front c . Express your answer in terms of q and γ_1 . **0.20**

B6 Calculate the speed of movement of the wave front during a detonation wave in a mixture of oxygen and hydrogen if the following chemical reaction occurs: **0.60**



The real process of hydrogen combustion is a set of reactions involving oxygen and hydrogen atoms, so ions H^+ and HO^- (this ion can be considered a diatomic molecule) must be included in the reaction products. In the mixture, there are 2 moles of hydrogen per mole of oxygen according to the reaction equation. For this reaction, assume $q \approx 15 \text{ MJ/kg}$ for every kilogram of hydrogen consumed.

Part C. Luminous detonation (3.0 points)

Under the influence of a strong electric field, the air is ionized and turns into plasma. A sufficiently strong field can be obtained by focusing laser radiation. The plasma, in turn, contains a large number of free charges and absorbs electromagnetic waves well, which leads to further heating of the plasma and propagation of the shock wave. On one side of the wave front there is plasma, on the other there is air. At the interface between air and plasma, laser radiation is absorbed and its energy is transferred to the plasma. Excessive pressure of heated air creates a shock wave propagating towards the laser radiation. This phenomenon is called a light detonation wave. Ramsden and Savich proposed to describe such a wave as a detonation wave, in which the analogue of the specific thermal effect of the reaction q is the value:

$$q_{sw} = \frac{I_0}{\rho_0 c},$$

where I_0 is the radiation intensity. In one of the first experiments observing a light detonation wave, a laser with power $P = 30 \text{ MW}$ was used, whose radiation was focused into a spot of

radius $r \approx 10^{-2}$ cm, creating an intensity:

$$I \approx \frac{P}{\pi r^2}.$$

The electric field near the focus of the lens is significantly greater than the field $E_{\text{probe}} = 30$ kV/cm causing air breakdown, so it is sufficient to form a light detonation wave.

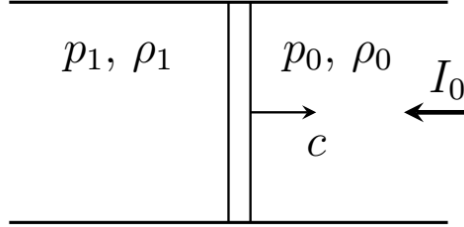


Figure 3: Figure 3.

Next, take the following data:

1. $p_0 = 1.0 \cdot 10^5$ Pa and $\rho_0 = 1.3$ kg/m³ (pressure and density of air at atmospheric pressure).
2. The plasma equation of state can be described by the equation of state of an ideal gas with the adiabatic exponent $\gamma_1 = 4/3$.
3. Work in the approximation of a strong wave, i.e., consider $p_0 \ll p_1$ and $\rho_0 c \ll I_0$.

C1	Obtain an equation for the light detonation adiabetic similar to point B2, that is, find the pressure p_1 in the plasma after the passage of the wave front. Express your answer in terms of $v_0, v_1, \gamma_1, I_0, c$.	1.20
C2	Determine the parameters (p_1 and v_1) of the Chapman-Jouguet point for the light-detonation adiabetic. Express your answer in terms of v_0, γ_1, I_0, c .	1.40
C3	Determine the speed of movement of the wave front c of the light detonation wave. Express your answer in terms of γ_1, I_0, ρ_0 . Calculate its numerical value.	0.40

Source: <https://pho.rs/p/3487>